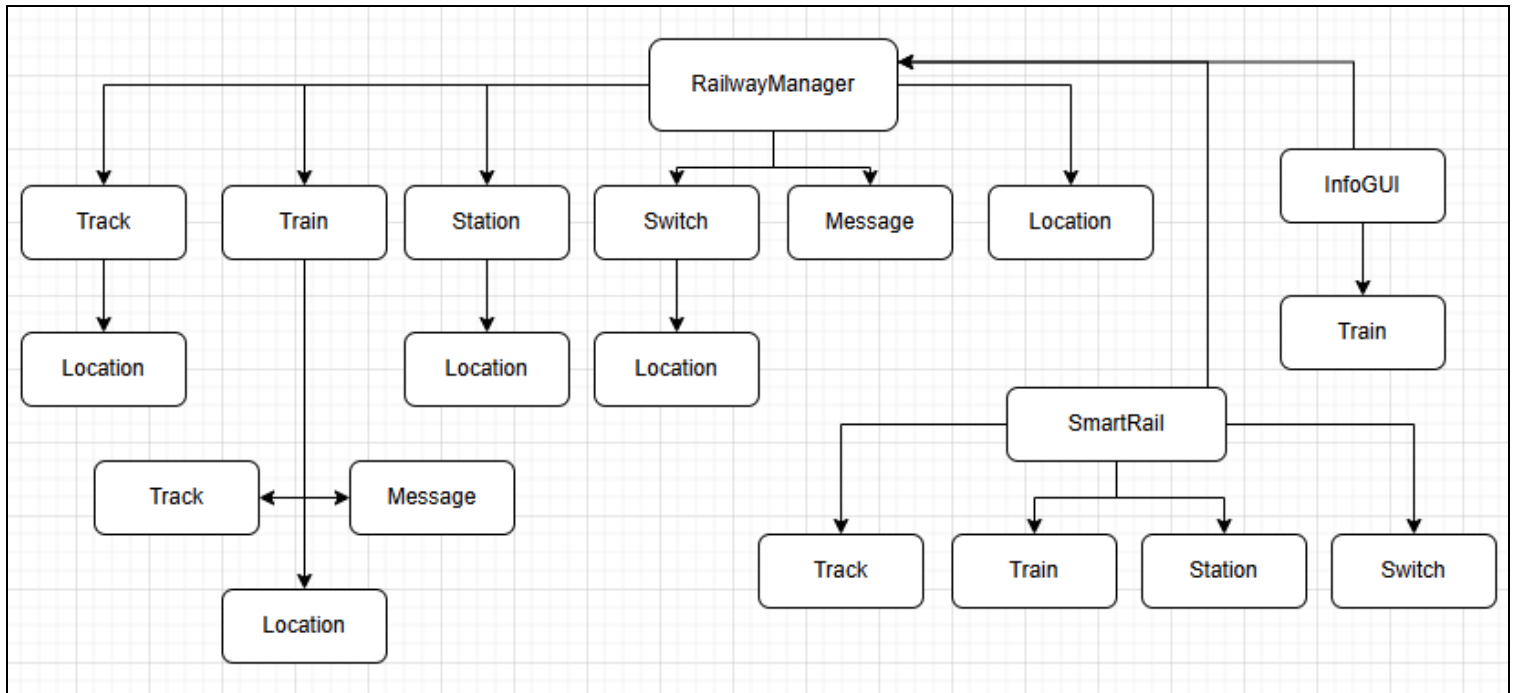




RailwayManager

The central controller of the program, managing Train, Track, Station, Switch, and Message objects. It facilitates pathfinding, locking, and unlocking tracks for trains and acts as a mediator for communication between components. RailwayManager references and manipulates almost every object to coordinate the simulation.

Train

Represents a train moving between Location objects, navigating across Track(s) and communicating with RailwayManager using Message objects. It updates its position, locks tracks, and unlocks them when finished. Train relies on the RailwayManager to provide valid paths.

Track

Represents a segment connecting two Location objects, used by Train for movement. Tracks can be (un)locked by Train through RailwayManager.

Station

A static point at a specific Location, representing the start or end of a train route. RailwayManager references Station for routing trains to and from specific Location coordinates. Stations do not interact directly with tracks or switches but act as destinations.

Switch

Represents a junction at a specific Location of tracks, allowing trains to change paths. RailwayManager controls switches, ensuring proper routing during pathfinding. Switch does not directly interact with trains but indirectly influences their paths.

Location

A coordinate object shared across Track, Station, Switch, and Train to represent positions. RailwayManager utilizes Location for pathfinding and managing component positions. Locations provide the spatial blueprint for all objects.

Message

A communication object passed between Train, RailwayManager, and other components. Used for actions like requesting locks, unlocking tracks, or confirming paths.

SmartRail and InfoGUI

GUI Components. SmartRail implements Listener/Observer pattern with the RailwayManager class to update the GUI visuals and serves as the entry point for the program. Uses basic JavaFX objects to represent the train simulation. InfoGUI takes in the user Route Requests and passes it to the SmartRail instance of the RailwayManager to run the simulation.